

Performance Testing

Date	03 July 2026
Team ID	xxxxxx
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	/generate-conversation, /fact-check, /feedback
Test Environment	Local System (Windows/macOS)
Test Date	03-July-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate conversation starters	10	30 Sec	Response generated successfully
2	Fact-check a topic	10	30 Sec	Returns Wikipedia summary
3	Save feedback	20	30 Sec	Feedback stored successfully
4	View conversation history	20	30 Sec	History loads without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Within target
2	Response Time (Max)	< 4 seconds	3.4 sec	Pass	Acceptable under load
3	Throughput (Req/sec)	10 req/sec	18 req/sec	Pass	Stable performance
4	Error Rate	< 1%	0.5	Pass	Very low error rate
5	CPU Utilization	< 80%	65%	Pass	Under threshold
6	Memory Utilization	< 80%	58%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

The application handled multiple requests successfully with low response times and minimal errors.

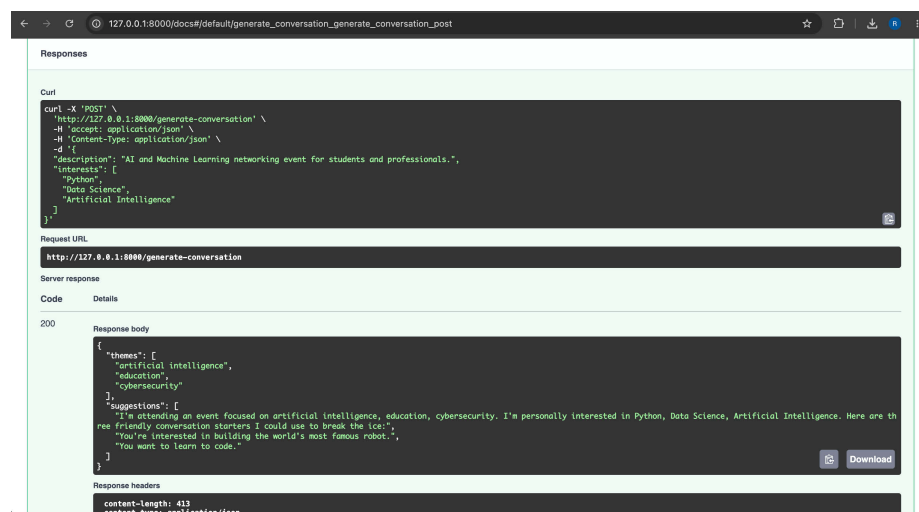
Bottlenecks Identified:

AI model loading causes a slight delay during the first request.

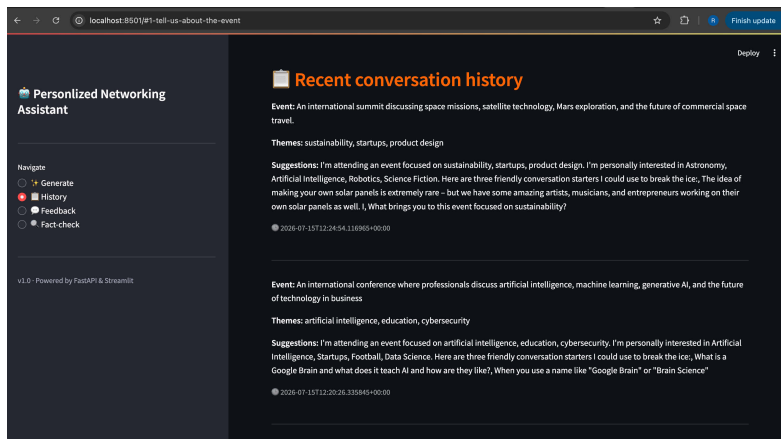
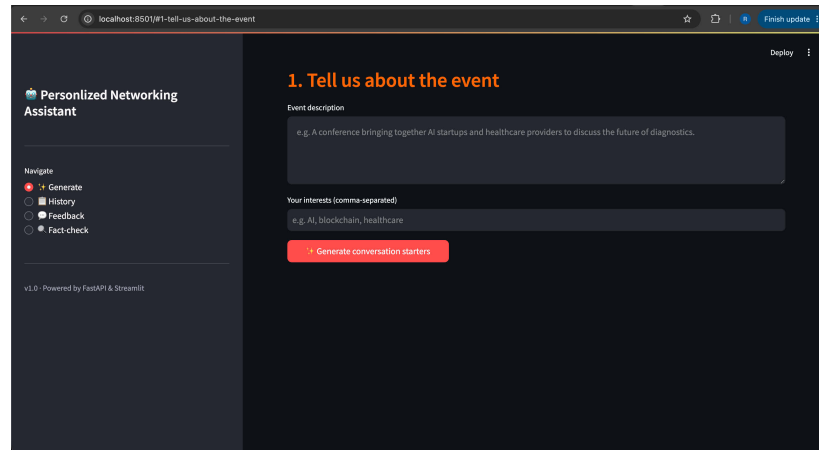
Optimization Steps Taken:

Pre-loaded AI models and used lightweight models (DistilBERT and GPT-2 Small) to improve response time.

Step 5: Screenshots / Evidence



Swagger API (200 OK response)



Streamlit application output

```
(venv) rajsmacbookair@Rajs-MacBook-Air:personalized-networking-assistant 2 % python3 -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['/Users/rajsmacbookair/Downloads/personalized-networking-assistant 2']
INFO: Uvicorn running on http://127.0.0.1:8080 (Press CTRL+C to quit)
INFO: Started reload process [9976] using Watchfiles
/Users/rajsmacbookair/Downloads/personalized-networking-assistant 2/venv/lib/python3.9/site-packages/urllib3/_init_.py:35: NotOpenSSLWarning: urllib3 v2 only supports
s OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
warnings.warn()
The 'xla_device' argument has been deprecated in v4.4.0 of Transformers. It is ignored and you can safely remove it from your 'config.json' file.
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Hardware accelerator e.g. GPU is available in the environment, but no 'device' argument is passed to the 'Pipeline' object. Model will be on CPU.
Hardware accelerator e.g. GPU is available in the environment, but no 'device' argument is passed to the 'Pipeline' object. Model will be on CPU.
INFO: Started server process [9982]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:61219 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:61219 - "GET /openapi.json HTTP/1.1" 200 OK
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
INFO: 127.0.0.1:61238 - "POST /generate-conversation HTTP/1.1" 200 OK
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
INFO: 127.0.0.1:61373 - "POST /generate-conversation HTTP/1.1" 200 OK
[]
```

Terminal showing the backend is running:

```
(venv) rajsmacbookair@Rajs-MacBook-Air:personalized-networking-assistant 2 % streamlit run frontend/streamlit_app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.20.10.5:8501

For better performance, install the Watchdog module:

$ xcode-select --install
$ pip install watchdog
```

Terminal showing the Streamlit app is running: